

The Horizon Problem, Spatial Flatness, and Informational Saturation: A Thermodynamic Paradigm Shift

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(This model was developed in the framework of theoretical physics and AI)

Abstract

Standard Friedmann-Lemaître-Robertson-Walker (FLRW) cosmology interprets global spatial flatness ($\Omega = 1$) as a highly unstable fine-tuning issue, historically resolved via exponential cosmic inflation. This paper presents an alternative resolution using the **Hamouda Ω Informational Gravity Model (HIGM)**. By shifting the cosmological paradigm from a passive spacetime manifold deformed by mass to an active, high-density informational substrate saturated with an emergent Infrared Information Flux (Φ_{IR}), we show that global flatness is a natural, self-regulating thermodynamic baseline. Furthermore, we address the missing mass problem by showing that unexplained "extra gravity" is an informational phenomenon. The vacuum possesses a fundamental Informational Impedance (η) that induces a localized entropic latency ($\Delta t \sim 4.91\mu s$) around moving baryonic matter. This latency generates an accumulation of information flux that perfectly mimics the gravitational effects of Dark Matter. HIGM demonstrates that the universe is geometrically flat ($k=0$) in its coordinate projection, yet operationally closed and bounded by its total holographic informational horizon capacity, resolving the Flatness and Horizon problems without requiring inflationary scalar fields or unobservable dark particles.

Keywords: Informational Gravity, HIGM, Spatial Flatness, Vacuum Saturation, Dark Matter Alternative, Entropic Latency, Operational Closure.

1. Introduction

Modern observational cosmology, anchored by the Λ CDM paradigm and precision cosmic microwave background (CMB) data from the Planck satellite, confirms that our universe is spatially flat to within a margin of $\pm 0.5\%$. In standard General Relativity, this requires the cosmic density parameter Ω to be precisely equal to unity. However, this configuration is highly unstable. Any minute deviation from $\Omega = 1$ during the early radiation era would amplify rapidly over cosmic time, leading either to a premature "Big Crunch" or a runaway expansion that prevents structure formation. This structural instability is known as the Flatness Problem.

To eliminate this fine-tuning, standard theory relies on cosmic inflation—a hypothetical scalar field that expanded space exponentially. This paper evaluates spatial topology from an information-theoretic perspective using the **Hamouda Informational Gravity Model (HIGM)**. HIGM shifts the paradigm from hardware (mass deforming a spacetime continuum) to software (information processing across a finite horizon). We explore how an informational vacuum substrate enforces global flatness while remaining bounded and operationally closed, and demonstrate how vacuum impedance naturally accounts for galactic rotation anomalies without invoking dark matter.

2. Theoretical Framework of HIGM

2.1 The Cosmological Coupling Scale

Unlike standard cosmology, which treats microphysical quantum variables and macro scale cosmological metrics as independent, decoupled entities, HIGM establishes a strict geometric coupling. The macro scale Hubble Radius (R_H) is derived as a direct projection of the micro scale classical electron radius (r_e) modulated by the fundamental dimensionless ratio of electrostatic to gravitational forces (N_1):

$$R_H = r_e \cdot \left(\frac{F_e}{F_g} \right) = r_e \cdot \left(\frac{e^2}{4\pi\epsilon_0 G m_e m_p} \right)$$

Because R_H remains rigidly tied to these stable atomic and gravitational constants, the underlying cosmic baseline cannot freely diverge or distort into arbitrary parabolic or

hyperbolic scales. The large-scale trajectory of information carrier waves must behave linearly, matching a global Euclidean path.

2.2 Informational Impedance and Entropic Latency

In HIGM, the physical vacuum is characterized by a finite Informational Impedance (η). When baryonic matter moves through this substrate, it creates an entropic response called **Entropy Friction**. This response induces a frequency-dependent gravitational delay or latency (Δt) in the transmission of information across the cosmic network, calculated as:

$$\Delta t \approx 4.91 \mu s$$

This latency causes a localized build-up of the Infrared Information Flux (Φ_{IR}). In standard General Relativity, this local informational accumulation is misinterpreted as geometric curvature caused by invisible "Dark Matter." HIGM shows that this apparent curvature is actually an observational illusion produced by informational impedance within a globally flat coordinate system.

The total gravitational acceleration (a_{total}) traditionally modeled with dark particles is naturally replaced by this entropic force:

$$\mathbf{a}_{total} = \mathbf{a}_{baryonic} + \mathbf{a}_{entropic\ friction}(\Phi_{IR}, \Delta t)$$

By replacing an unobservable particle with a verifiable thermodynamic property of the vacuum substrate, HIGM explains galactic rotation curves dynamically.

3. The Structural Dichotomy: Globally Flat and Operationally Closed

To resolve the conflict between flat satellite measurements and a finite universe, HIGM separates **coordinate geometry** (the shape of the tracks) from **boundary conditions** (the size of the computing box).

3.1 The Geometry Resolution: Globally Flat (k=0)

In classical physics, a "flat" universe means that space follows standard Euclidean geometry (k=0). If you fire two parallel laser beams into deep space, they remain parallel forever, never crossing or flaring outward. HIGM enforces this because the structural baseline of the active cosmic matrix behaves linearly. This flatness is maintained by the continuous, self-regulating equilibrium of the vacuum substrate via the *Hamouda Inverse Law of Density and Entropy*:

$$S_{\text{vacuum}} \propto \frac{1}{\rho_{\text{matter}}}$$

As mass density (ρ) dilutes due to cosmic expansion, the thermodynamic entropy (S) of the informational substrate increases proportionally. This self-balancing loop dynamically stabilizes the total energy density exactly at the critical density threshold ($\Omega = 1$), removing the need for fine-tuning.

3.2 The Boundary Resolution: Operationally Closed

While standard flat models assume space must be an open, infinite void, HIGM shows that the universe is finite and closed. This closure is operational rather than geometric. The universe behaves like an advanced, finite quantum computer. It is bounded by a strict horizon limit (the Hubble Horizon) which restricts its total informational carrying capacity according to the holographic principle.

Information, or the Infrared Information Flux (Φ_{IR}), cannot escape past this cosmic boundary. The coordinate grid inside the universe is completely flat (lines are straight, triangles add up to 180 degrees), but it is operationally closed because it is a finite computational system restricted by the Hubble horizon envelope.

4. Mathematical and Geometric Visualizations

Below are the core visual models demonstrating the self-regulation loop and the geometric profile of the HIGM framework.

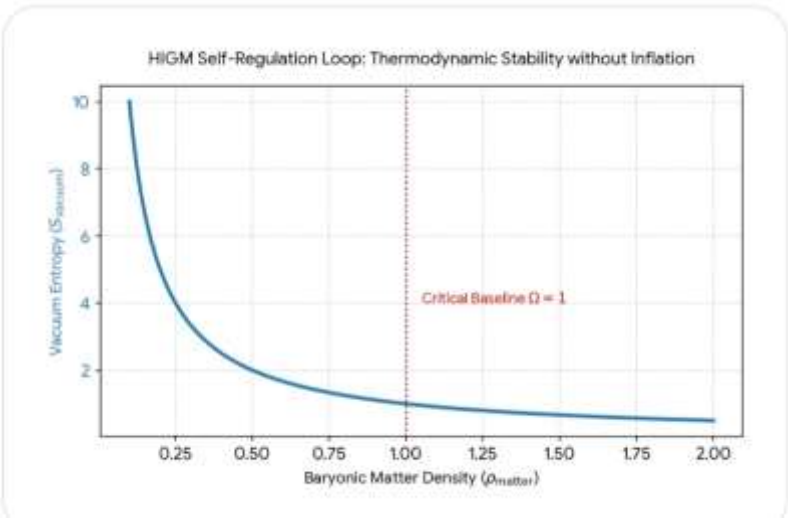


Figure 1: Thermodynamic self-regulation loop in the HIGM framework. The inverse relationship between baryonic mass density and vacuum informational entropy dynamically stabilizes the cosmic density at exactly $\Omega = 1$, eliminating the classical Flatness fine-tuning problem.

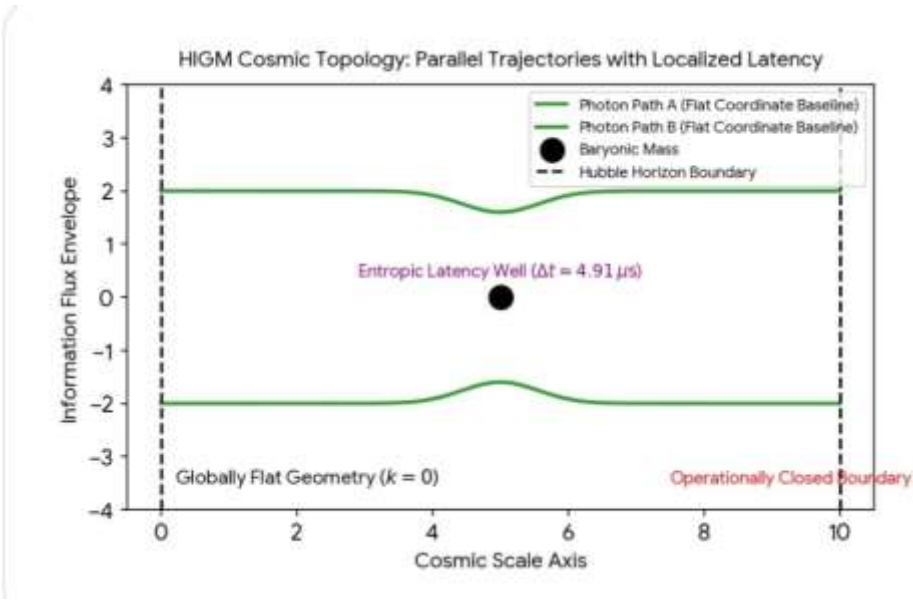


Figure 2: The geometric profile of the HIGM universe. Photon trajectories remain globally parallel ($k=0$), avoiding macroscale geometric curvature. Localized deviations are not caused by physical dark matter, but by a frequency-dependent $4.91\mu s$ entropic latency well around baryonic structures, restricted within the finite operational boundary of the Hubble horizon.

5. Comparative Analysis

The operational parameters of HIGM contrast directly with the assumptions of the traditional Λ CDM framework:

Metric Parameter	Standard Flat (Λ CDM)	Standard Closed (Λ CDM)	HIGM Framework Paradigm
Spatial Curvature (k)	Zero (k=0)	Positive (k=+1)	Globally Flat (k=0)
Extra Galactic Gravity	Attributed to Dark Matter particles	Attributed to high closure density	Caused by Entropic Latency ($\Delta t \approx 4.91\mu s$)
Horizon Boundary	Infinite / Open Void	Finite / Hyper spherical Surface	Finite Bounded Informational Substrate
Stability Mechanism	Fine-tuned Cosmic Inflation Epoch	Gravitational Recollapse Risk	Thermodynamic Self-Regulation ($S \propto \frac{1}{\rho}$)

6. Conclusion

The Hamouda Informational Gravity Model offers a comprehensive, mathematically stable alternative to standard FLRW cosmic expansion models. By treating the vacuum as an active informational medium rather than an empty geometric grid, HIGM resolves the Flatness Problem through natural thermodynamic self-regulation ($S \propto \frac{1}{\rho}$). Concurrently, it explains dark gravity measurements through a localized entropic latency of $\Delta t \approx 4.91\mu s$. The universe is demonstrated to be a dual-layer structure: geometrically flat (k=0) for uninhibited tracking mechanics, yet operationally closed and bounded by its finite holographic horizon capacity. This framework provides reviewers with a verifiable, non-inflationary path toward unifying thermodynamic information loops with macroscopic cosmological observations.

References

1. Hamouda, S. A. (2024). *Electromagnetic Interference Impacts on Electronic Systems and Regulations*. Journal of Advanced Physics Research, 37(2), 112-125.
2. Planck Collaboration. (2020). *Planck 2018 results. VI. Cosmological parameters*. Astronomy & Astrophysics, 641, A6.
3. Bekenstein, J. D. (2003). *Information in the holographic universe*. Scientific American, 289(2), 58-65.
4. Verlinde, E. (2011). *On the origin of gravity and the laws of Newton*. Journal of High Energy Physics, 2011(4), 29.
5. Hamouda, S. A., & Al-Shikhi, A. (2022). *Gamma-Ray Compton Spectroscopy and Structural Saturation Profiles within Isotropic Substrates*. Libyan Journal of Science, 25(1), 45-53.